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Education

H.B.Sc., Mathematics and Statistics — 2018-2022
University of Toronto, Toronto, ON

Technical Skills

Data: Python, Numpy, Pandas, PyTorch, Scikit-Learn, Matplotlib, Seaborn, SQL, NoSQL, R, Excel
Other: shell scripting, vim, version control (git), cloud computing (AWS, Azure, Google Cloud Platform), Unix/Linux, Astropy, Selenium, NLTK, LaTeX, regular expressions

Relevant Coursework

2019 Fall	Stars and Planets
2019 Fall	Probability and Statistics I
2019 Fall	Ordinary Differential Equations
2019 Summer	Multivariable Calculus
2019 Winter	Introduction to Computer Science
2018-2019	Calculus
2018-2019	Linear Algebra I, II
2018 Fall	An Introduction to Statistical Reasoning and Data Science

Independent Coursework

2019 **Data-Driven Astronomy**, Coursera
Pulsars, black holes, exoplanets, stellar evolution, galaxies. Image stacking, crossmatching, data management, regression, classification. Implementation in Python with Astropy, and SQL.

2019 **Intro to Deep Learning with PyTorch**, Udacity
Mathematics of neural networks. Data preparation. Image classification with CNNs and OpenCV. Time series with RNNs. Implementation in Python with PyTorch.

2018 **Intro to Machine Learning**, Udacity
Classifiers (naïve Bayes, SVMs, ensemble methods), data processing, regression, clustering, outlier detection, PCA, evaluation metrics. Implementation in Python with sklearn.

Volunteering

2015-2018 **Python Workshop Lead**, Sir Winston Churchill Secondary, Vancouver, B.C.
Coordinated and delivered workshops to prepare students for a national programming contest.

Projects

2019 **Image classification with CNNs**
Cleaned a dataset of several thousands of images of flowers. Optimized a deep learning model and applied cutting-edge techniques to achieve 99% accuracy in the classification of species of flowers. Could potentially be adapted for classifcation of astronomical objects.

2018 **Analysis of global Internet usage statistics**
Spearheaded the programming aspect of a collaborative statistical analysis project. Proposed research questions and used R to develop data-backed answers. Gave a presentation of results.

Interests

Languages: English (native), French, Chinese, Norwegian (conversational)
Other: *nix customization, urban beekeeping, baking, reading, computer programming